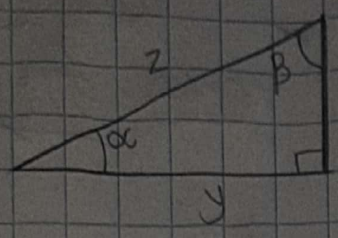
Tarea Lección 5.2

7)



a) $\alpha = B = 0.28$

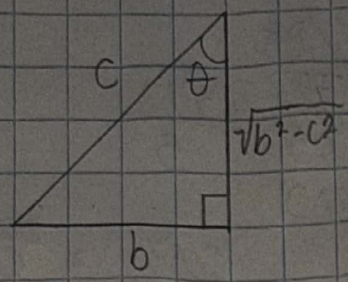
b) $\beta = D = 1.29$

c) $x = A = 7$

d) $y = C = 24$

e) $z = E = 25$

9)



$$b^2 + a^2 = c^2$$
$$a^2 = c^2 - b^2$$
$$a = \sqrt{b^2 - c^2}$$

$\operatorname{sen}\theta = \dfrac{b}{c}$

$\cos\theta = \dfrac{\sqrt{b^2-c^2}}{c}$

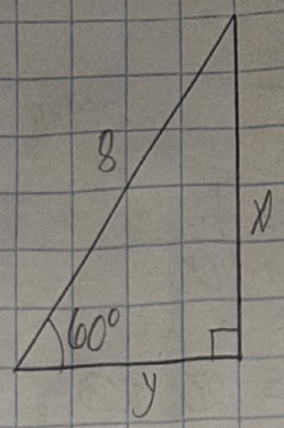
$\tan\theta = \dfrac{b}{\sqrt{b^2-c^2}}$

$\cot\theta = \dfrac{\sqrt{b^2-c^2}}{b}$

$\sec\theta = \dfrac{c}{\sqrt{b^2-c^2}}$

$\csc\theta = \dfrac{c}{b}$

15)



$\operatorname{sen}60° = \dfrac{\sqrt{3}}{2}$ $\qquad$ $\operatorname{sen}60° = \dfrac{x}{8}$

$$\frac{\sqrt{3}}{2} = \frac{x}{8}$$
$$8\sqrt{3} = 2x$$
$$x = 4\sqrt{3}$$

$\cos 60° = \dfrac{1}{2}$ $\qquad$ $\cos 60° = \dfrac{y}{8}$

$$\frac{1}{2} = \frac{y}{8}$$
$$8 = 2y$$
$$y = 4$$

21) $\sec\theta = \frac{6}{5}$ $\cos\theta = \frac{5}{6}$ $\text{sen}\,\theta = \frac{\sqrt{77}}{6}$

$\csc = \frac{6}{\sqrt{77}}$ $\tan\theta = \frac{\sqrt{77}}{5}$ $\cot\theta = \frac{5}{\sqrt{77}}$

$\text{sen}^2\theta + \left(\frac{5}{6}\right)^2 = 1$

$\text{sen}^2\theta = 1 - 25/36$

$\text{sen}^2\theta = \frac{11}{36}$

$\text{sen}\,\theta = \sqrt{\frac{77}{36}} = \frac{\sqrt{77}}{6}$

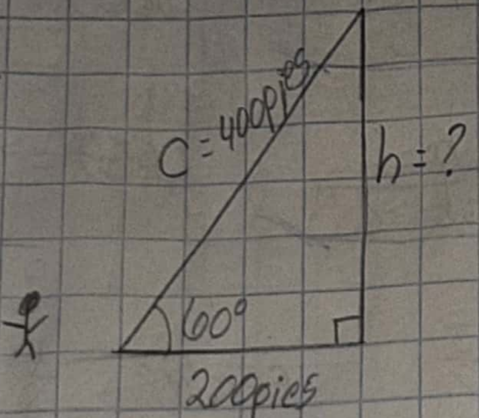
$\tan^2\theta + 1 = \left(\frac{6}{5}\right)^2$

$\tan^2\theta = \left(\frac{6}{5}\right)^2 - 1$

$\tan^2\theta = 11/25$

$\tan\theta = \sqrt{\frac{77}{25}} = \frac{\sqrt{77}}{5}$

23)



$\cos 60° = \frac{1}{2}$ $\cos 60° = \frac{200}{c}$

$\frac{1}{2} = \frac{200}{c}$

$c = 400$

$\text{sen}\,60° = \frac{\sqrt{3}}{2}$ $\text{sen}\,60° = \frac{h}{400}$

$\frac{\sqrt{3}}{2} = \frac{h}{400}$

$400\sqrt{3} = 2h$

$h = 200\sqrt{3}$

La altura del árbol es

$200\sqrt{3} \approx 346.4$ pies

27) $\frac{(1.22)(550 \times 10^{-9})}{\text{sen}\,0.0003769} = 1.02\,m$

33) a) $\text{Sen } 30^\circ = \underline{0.5}$ c) $\cos \pi^\circ = \underline{0.9985}$

b) $\sec(2\pi/5) = \underline{-0.9880}$ d) $\cos \pi = \underline{-1}$

37) a) $5\text{sen}^2\theta + 5\cos^2\theta$

$5(\text{sen}^2\theta + \cos^2\theta) = 5(1) = \underline{5}$

b) $5\text{sen}^2(\theta/4) + 5\cos^2(\theta/4)$

$5(\text{sen}^2(\theta/4) + \cos^2(\theta/4))$

$5(1) = \underline{5}$

43) $\dfrac{2 - \tan\theta}{2\csc\theta - \sec\theta}$

$$\frac{2 - \dfrac{\text{sen}\theta}{\cos\theta}}{2\left(\dfrac{1}{\text{sen}\theta}\right) - \dfrac{1}{\cos\theta}} = \frac{\dfrac{\cancel{2\cos\theta - \text{sen}\theta}}{\cancel{\cos\theta}}}{\dfrac{\cancel{2\cos\theta - \text{sen}\theta}}{\text{sen}\theta \cancel{\cos\theta}}} = \underline{\text{Sen}\theta}$$

63) $(1+\operatorname{Sen}\theta)(1-\operatorname{Sen}\theta) = \frac{1}{\sec^2\theta}$

$1-\operatorname{sen}^2\theta = \cos^2\theta$

$\cos^2\theta = \frac{1}{\sec^2\theta}$

65) $\sec\theta - \cos\theta = \tan\theta \operatorname{sen}\theta$

$\frac{1}{\cos\theta} - \cos\theta = \frac{1-\cos^2\theta}{\cos\theta}$

$1-\cos^2\theta = \frac{\operatorname{sen}^2\theta}{\cos\theta} = \tan\theta$

$\frac{\operatorname{sen}^2\theta}{\cos\theta} \cancel{*} \operatorname{sen}\theta = \tan\theta \operatorname{sen}\theta$

69) $(\cot\theta + \csc\theta)(\tan\theta - \operatorname{sen}\theta) = \sec\theta - \cos\theta$

$\cot\theta \tan\theta - \cot\theta \operatorname{sen}\theta + \csc\theta \tan\theta - \csc\theta \operatorname{sen}\theta$

$\frac{1}{\tan t} \tan\theta - \frac{\cos\theta}{\operatorname{sen}\theta} \operatorname{sen}\theta + \frac{1}{\operatorname{sen}\theta}\left(\frac{\operatorname{sen}\theta}{\cos\theta}\right) - \frac{1}{\operatorname{sen}\theta} \operatorname{Sen}\theta$

$1 - \cos\theta + \frac{1}{\cos\theta} - 1$

$-\cos\theta + \sec\theta$

73) $\log \csc\theta = -\log \operatorname{sen}\theta$

$$\log\left(\frac{1}{\operatorname{sen}\theta}\right) = \log 1 - \log \operatorname{sen}\theta = -\log \operatorname{sen}\theta$$

$$= 0$$

77) $P(-2, -5)$

$$r = \sqrt{(-2)^2 + (-5)^2} = \sqrt{4+25} = \sqrt{29}$$

$\operatorname{sen}\theta = \dfrac{-5}{\sqrt{29}}$ $\qquad$ $\csc\theta = \dfrac{\sqrt{29}}{-5}$

$\cos\theta = \dfrac{-2}{\sqrt{29}}$ $\qquad$ $\sec\theta = \dfrac{\sqrt{29}}{-2}$

$\tan\theta = \dfrac{5}{2}$ $\qquad$ $\cot\theta = \dfrac{2}{5}$

Escaneado con CamScanner

85) a) $90°$ b) $0°$

$\text{sen}\,90° = 1$ $\cot 90° = 0$
$\cos 90° = 0$ $\sec 90° = /$
$\tan 90° = /$ $\csc 90° = 1$

$\text{sen}\,0° = 0$ $\cot 0° = /$
$\cos 0° = 1$ $\sec 0° = 1$
$\tan 0° = 0$ $\csc 90° = /$

c) $7\pi/2$ d) 3π

$\text{sen}\,7\pi/2 = -1$ $\cot 7\pi/2 = 0$
$\cos 7\pi/2 = 0$ $\sec 7\pi/2 = /$
$\tan 7\pi/2 = /$ $\csc 7\pi/2 = -1$

$\text{sen}\,3\pi = 0$ $\cot 3\pi = /$
$\cos 3\pi = -1$ $\sec 3\pi = -1$
$\tan 3\pi = 0$ $\csc 3\pi = /$

87) a) $\cos\theta > 0$ y $\text{sen}\,\theta < 0$

$\cos\theta\,(+)$ $\text{sen}\,\theta\,(-)$ $= + * - = - =$ IV

b) $\text{sen}\,\theta < 0$ y $\cot\theta > 0$

$\text{sen}\,\theta\,(-)$ $\cot(+)$ $= - * + = - =$ III

c) $\csc\theta > 0$ y $\sec\theta < =$

$\csc\theta\,(+)$ $\sec\theta\,(+)$ $= + * + = + =$ II

d) $\sec\theta < 0$ y $\tan\theta > 0$

$\sec\theta\,(-)$ $\tan\theta\,(+)$ $=$ III

Escaneado con CamScanner

101) $\sqrt{sen^2(\theta/2)}$ $2\pi < \theta < 4\pi$

$\cancel{\sqrt{sen^{\cancel{2}}}} = -sen \frac{\theta}{2}$